

Laboratorium
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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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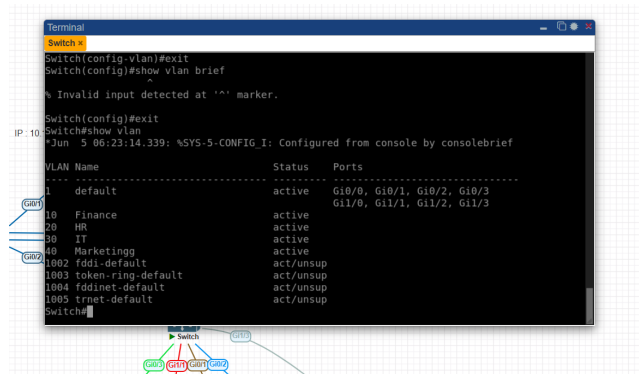
2026

1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke mode konfigurasi Cisco Switch menggunakan perintah:

```
enable
configure terminal
```

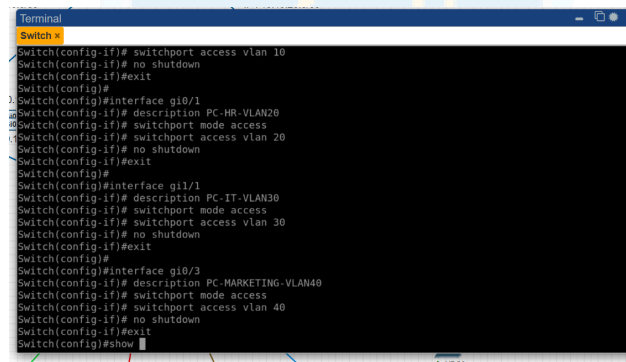


2. Membuat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing).

3. Melakukan verifikasi VLAN menggunakan perintah:

```
show vlan brief
```

4. Mengkonfigurasi port yang terhubung ke PC sebagai access port sesuai VLAN masing-masing.



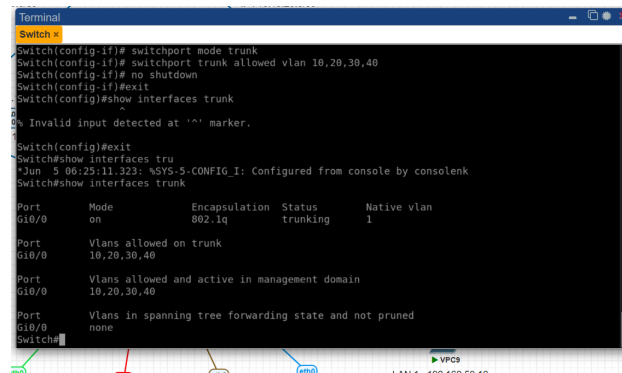
5. Mengkonfigurasi interface yang terhubung ke Cisco Router sebagai trunk port.
6. Mengizinkan VLAN 10, 20, 30, dan 40 melewati trunk menggunakan konfigurasi allowed VLAN.
7. Melakukan verifikasi trunk dengan perintah:

```
show interfaces trunk
```

8. Menyimpan konfigurasi menggunakan perintah:

```
write memory
```

9. Memastikan VLAN dan trunk telah aktif sesuai topologi yang digunakan.



```
Switch>
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30,40
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# show interfaces trunk
% Invalid input detected at '^' marker.
Switch(config)# exit
Switch# show interfaces tru
Jun  5 06:25:11.323: %SYS-5-CONFIG_I: Configured from console by consolenk
Switch# show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
-----
Gi0/0/0   on        802.1q          trunking      1

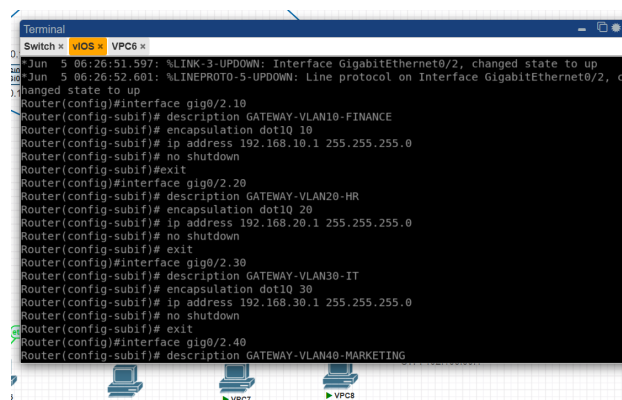
Port      Vlans allowed on trunk
-----
Gi0/0/0   10,20,30,40

Port      Vlans allowed and active in management domain
-----
Gi0/0/0   10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
-----
Gi0/0/0   none
Switch#
```

1.2 Praktikum 2: Konfigurasi Cisco Router

1. Masuk ke mode konfigurasi Cisco Router.
2. Mengkonfigurasi interface yang terhubung ke Cisco Switch sebagai trunk (Router on a Stick).
3. Membuat subinterface VLAN 10, VLAN 20, VLAN 30, dan VLAN 40.
4. Memberikan alamat IP gateway pada masing-masing subinterface.



```
Router>
Router(config)# interface gi0/2.10
Router(config-subif)# description GATEWAY-VLAN10-FINANCE
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)# interface gi0/2.20
Router(config-subif)# description GATEWAY-VLAN20-HR
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)# interface gi0/2.30
Router(config-subif)# description GATEWAY-VLAN30-IT
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)# interface gi0/2.40
Router(config-subif)# description GATEWAY-VLAN40-MARKETING
```

5. Mengkonfigurasi DHCP Server untuk VLAN 10 dan VLAN 20.
6. Mengatur alamat IP statis pada PC VLAN 30 dan VLAN 40.
7. Mengatur PC VLAN 10 dan VLAN 20 agar memperoleh IP secara DHCP.
8. Melakukan verifikasi interface menggunakan:

`show ip interface brief`

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
Router#
*Jun  5 06:30:53.710: %SYS-5-CONFIG-I: Configured from console by consoleshow
% Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface      IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0      unassigned      YES unset  administratively down  down
GigabitEthernet0/1      unassigned      YES unset  administratively down  down
GigabitEthernet0/2      unassigned      YES unset  up                  up
GigabitEthernet0/2.10   192.168.10.1    YES manual  up                  up
GigabitEthernet0/2.20   192.168.20.1    YES manual  up                  up
GigabitEthernet0/2.30   192.168.30.1    YES manual  up                  up
GigabitEthernet0/2.40   192.168.40.1    YES manual  up                  up
GigabitEthernet0/3      unassigned      YES unset  administratively down  down
GigabitEthernet0/4      unassigned      YES unset  administratively down  down
GigabitEthernet0/5      unassigned      YES unset  administratively down  down
Router#

```

9. Melakukan verifikasi DHCP Pool menggunakan:

show ip dhcp pool

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
10 Router#show ip dhcp pool
Pending event      : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.10.12      192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.20.12      192.168.20.1 - 192.168.20.254  1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.8d Jun 06 2026 06:30 AM Automatic
192.168.20.11    0100.5079.6668.8c Jun 06 2026 06:30 AM Automatic
Router#

```

10. Melakukan verifikasi DHCP Binding menggunakan:

show ip dhcp binding

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
10 Router#show ip dhcp binding
Pending event      : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.10.12      192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
192.168.20.12      192.168.20.1 - 192.168.20.254  1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.8d Jun 06 2026 06:30 AM Automatic
192.168.20.11    0100.5079.6668.8c Jun 06 2026 06:30 AM Automatic
Router#

```

11. Menguji konektivitas dengan melakukan ping ke gateway masing-masing VLAN.

```

Terminal
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 >
0 DDORA IP 192.168.10.11/24 GW 192.168.10.1
3 VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86396, 86400/43200/75600
MAC        : 00:50:79:66:68:8d
I/POR      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.349 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.441 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.866 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.238 ms
VPCS>

```



```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x
ODORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.20.1/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86398, 86400/43200/75600
MAC        : 00:50:79:66:68:8c
LPORT      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=4.099 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=2.122 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.429 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.007 ms
^C
VPCS>

```

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 00:50:79:66:68:8b
LPORT      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.924 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.540 ms
^C
VPCS> [23-]

```

12. Menguji komunikasi antar VLAN untuk memastikan inter-VLAN routing berjalan dengan baik.

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.238 ms
^C
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.083 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.483 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.557 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.623 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=2.163 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=2.921 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=3.802 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.604 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.401 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.408 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.628 ms
^C
VPCS>

```

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

1. Mengkonfigurasi interface LAN pada Huawei Router dengan jaringan 192.168.50.0/24.
2. Mengkonfigurasi alamat IP pada PC LAN Huawei.

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team.
Press '?' to get help.

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:8e
LPORT      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS>

```

3. Melakukan pengujian konektivitas antara PC LAN Huawei dengan gateway Huawei.

```

Switch > VPC5 > VPC6 > VPC5 > VPC7 > NE40E > VPC9 >
VPC5> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1
VPC5> show ip
NAME      : VPC5[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:0e
PORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500
VPC5> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=5.759 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.334 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.053 ms
^C
VPC5>

```

4. Mengkonfigurasi alamat IP pada link Cisco Router dan Huawei Router Link 1 (10.10.10.0/30).

5. Mengkonfigurasi alamat IP pada link Cisco Router dan Huawei Router Link 2 (10.10.11.0/30).

6. Mengkonfigurasi alamat IP pada link Huawei Router dan MikroTik (10.10.20.0/30).

7. Mengkonfigurasi alamat IP pada link Cisco Router dan MikroTik (10.10.30.0/30).

8. Melakukan verifikasi interface pada Cisco Router, Huawei Router, dan MikroTik Router.

```

Switch > VPC5 > VPC6 > VPC7 > NE40E > VPC9 > MikroTik >
Router(config)#interface gig0/1
Router(config-if)#description LINK 10-R00001A
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Jun  5 06:30:58 499: NLINK-1-UPDOWN: Interface GigabitEthernet0/1, changed state to up
Jun  5 06:30:51 501: NLINK-1-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#exit
Router#show
Jun  5 06:40:17 803: SYS-5-COMF0: Configured from console by console
Type "show" for a list of subcommands
Router#show ip interface brief
Interface      IP-Address      OK? Method Status    Protocol
GigabitEthernet0/0  10.10.11.1      YES manual  up        up
GigabitEthernet0/1  10.10.10.1      YES manual  up        up
GigabitEthernet0/2  unassigned      YES unset   up        up
GigabitEthernet0/2.10  10.10.10.1      YES manual  up        up
GigabitEthernet0/2.20  10.10.20.1      YES manual  up        up
GigabitEthernet0/2.30  10.10.30.1      YES manual  up        up
GigabitEthernet0/2.40  10.10.40.1      YES manual  up        up
GigabitEthernet0/3  10.10.10.1      YES manual  up        up
GigabitEthernet0/4  unassigned      YES unset   administratively down down
GigabitEthernet0/5  unassigned      YES unset   administratively down down
Router#

```

```

Switch > VPC5 > VPC6 > VPC7 > NE40E > VPC9 > MikroTik >
MikroTik>display ip interface brief
down: administratively down
down: FEB overload down
down: standby
(!): loopback
(!): spoofing
(!): Dumping suppressed
(!): E-Trunk down
The number of interface that is up in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is up in Protocol is 14
The number of interface that is DOWN in Protocol is 7
Interface      IP Address/Mask  Physical  Protocol VPN
ethermt0/0/0    10.10.10.2/20    up        up        -
ethermt0/0/1    128.1.130.137/10 up        down      -
ethermt0/0/2    128.1.130.137/10 up        up        13vpn
ethermt0/0/3    192.168.50.1/24  up        up        -
ethermt0/0/4    128.1.130.137/10 up        up        13vpn
ethermt0/0/5    10.10.20.1/20    up        up        -
ethermt0/0/6    128.1.130.137/10 up        up        13vpn
ethermt0/0/7    10.10.11.2/20    up        up        -
ethermt0/0/8    128.1.130.137/10 up        up        13vpn
---- More ----

```

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <

[admin@Mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@Mikrotik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.10.20.2/30 10.10.20.0 ether1
1 10.10.30.2/30 10.10.30.0 ether2
[admin@Mikrotik] >
```

9. Melakukan pengujian ping antar-router untuk memastikan seluruh link aktif dan dapat saling berkomunikasi.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <

Router(config)#exit
Router#
*Jun  5 06:41:12.127: %SYS-5-CONFIG I: Configured from console by console
Router#display ip interface brief
% Invalid input detected at '^' marker.

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#
```

10. Memastikan seluruh router dapat melakukan ping ke router tetangga secara langsung.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <

--- 10.10.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms
```

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

[admin@Mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@Mikrotik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ::: TO-HUAWEI 10.10.20.0 ether1
1 ::: TO-CISCO 10.10.20.0 ether2
[admin@Mikrotik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 2ms
1 10.10.20.1 56 255 1ms
2 10.10.20.1 56 255 6ms
sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=3ms max-rtt=6ms
[admin@Mikrotik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 2ms
2 10.10.30.1 56 255 2ms
sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
[admin@Mikrotik] >

```

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengkonfigurasi OSPF pada Cisco Router dengan Router ID 1.1.1.1.
2. Mengiklankan seluruh network VLAN dan link antar-router ke area OSPF.
3. Mengatur nilai OSPF Cost pada interface Cisco Router sesuai skenario topologi.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Sending updates every 0 seconds
  Inetlink after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum paths 32
  Routing for Networks:
    Routing Information Sources:
      Gateway Distance Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Sending updates every 0 seconds
  Inetlink after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum paths 4
  Routing for Networks:
    10.10.0.0 0.0.0.3 area 0
    10.10.10.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway Distance Last Update
  Distance: (default is 110)

Router#

```

4. Mengkonfigurasi OSPF pada Huawei Router dengan Router ID 2.2.2.2.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

[*HUAWEI-Ethernet1/0/3] description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[*HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
        Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: None MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h00m12s
Neighbor Up Time : 2026-06-05 07:01:17
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: None MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m12s
Neighbor Up Time : 2026-06-05 07:01:17
Authentication Sequence: [ 0 ]

[*HUAWEI]

```

5. Mengiklankan seluruh network Huawei ke area OSPF.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ospf routing
OSPF Process 1 with Router ID 2.2.2.2
Routing Tables
Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0
Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
--- More ---
```

6. Mengatur nilai OSPF Cost pada interface Huawei Router.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9 Routes : 14
OSPF routing table status : <Active>
Destinations : 5 Routes : 10
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
OSPF 10 110 D 10.10.10.1 Ethernet1/0/4
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
```

7. Mengkonfigurasi OSPF pada MikroTik Router dengan Router ID 3.3.3.3.

8. Mengiklankan network yang terhubung ke MikroTik ke area backbone OSPF.

```
Switch < vIOS < VPC6 < VPC5 < VPC8 < VPC7 < NE40E < VPC9 < Mikrotik <
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add net
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add ne
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add n
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network add
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network a
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf network
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf netwo
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf netw
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf netv
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf net
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf ne
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf n
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf
[admin@mikrotik] > /routing ospf network add network=10.20.0/30 area=backbone/routing ospf
[admin@mikrotik] > /routing ospf interface add interface=ether1 cost=100
[admin@mikrotik] > /routing ospf interface add interface=ether2 cost=100
[admin@mikrotik] > /routing ospf neighbor print
[admin@mikrotik] > /routing ospf neighbor print
[admin@mikrotik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@mikrotik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.30.1 interface=ether2 priority=1
dr-address=10.30.1 backup-dr-address=0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=95
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=45
[admin@mikrotik] >
```

9. Mengatur nilai OSPF Cost pada interface MikroTik sebagai jalur cadangan.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routing ospf n
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routing ospf
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routing ospf
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print

[admin@MikroTik] > /routing ospf neighbor print

[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=9s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=4s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
b - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 Ado 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.30.1 110
[admin@MikroTik] >

```

10. Melakukan verifikasi OSPF Neighbor pada Cisco Router, Huawei Router, dan MikroTik Router.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
Routing for Network
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 0)

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 1.1.1.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.10.10.0 0.0.0.0 area 0
10.10.11.0 0.0.0.0 area 0
10.10.20.0 0.0.0.0 area 0
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 110)

Router#
Jun 5 06:47:14.143: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Jun 5 06:47:14.143: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
Jun 5 06:49:44.472: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/DR 00:00:36 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#

```

11. Melakukan verifikasi tabel routing OSPF pada seluruh router.

```

Switch x VIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x
OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 26 sec
Retrans timer interval: 5
Neighbor is up for 00h00m07s
Neighbor Up Time : 2026-06-05 07:01:17
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
--- More ---

```

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
[admin@Mikrotik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@Mikrotik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=9s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=4s
[admin@Mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS   PREF-SRC   GATEWAY      DISTANCE
0 Ado 10.10.10.0/30          10.10.20.1      110
1 Ado 10.10.11.0/30          10.10.20.1      110
2 Ado 192.168.10.0/24         10.10.30.1      110
3 Ado 192.168.20.0/24         10.10.30.1      110
4 Ado 192.168.30.0/24         10.10.30.1      110
5 Ado 192.168.40.0/24         10.10.30.1      110
6 Ado 192.168.50.0/24         10.10.20.1      110
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=1m46s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=1m41s
[admin@Mikrotik] > █
```

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
IA - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
0 10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:10, GigabitEthernet0/0
1 10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:10, GigabitEthernet0/3
0 192.168.50.0/24 [110/11] via 10.10.11.2, 00:04:10, GigabitEthernet0/0
1 192.168.50.0/24 [110/11] via 10.10.10.2, 00:04:10, GigabitEthernet0/3

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
IA - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
0 10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:30, GigabitEthernet0/0
1 10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:30, GigabitEthernet0/3
0 192.168.50.0/24 [110/11] via 10.10.11.2, 00:04:30, GigabitEthernet0/0
1 192.168.50.0/24 [110/11] via 10.10.10.2, 00:04:30, GigabitEthernet0/3

Router# █
```

12. Melakukan pengujian konektivitas end-to-end antara VLAN Cisco dan LAN Huawei menggunakan perintah ping.

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.110 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.291 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.684 ms
^C
VPCS> █
```

```
Switch > VIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.622 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.673 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=9.393 ms
^C
VPCS> █
```

```
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=5.808 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=5.901 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=4.087 ms
^C
VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.972 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=3.828 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=4.909 ms
^C
VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.018 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=5.734 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.692 ms
^C
VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=5.480 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=8.297 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=4.834 ms
^C
VPCS> 
```

13. Memastikan seluruh network telah dipelajari melalui OSPF dan dapat saling berkomunikasi.

```
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30          10.10.20.1    110
1 ADo 10.10.11.0/30         10.10.30.1    110
2 ADo 192.168.10.0/24        10.10.30.1    110
3 ADo 192.168.20.0/24        10.10.30.1    110
4 ADo 192.168.30.0/24        10.10.30.1    110
5 ADo 192.168.40.0/24        10.10.30.1    110
6 ADo 192.168.50.0/24        10.10.20.1    110
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=lm46s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=lm41s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30          10.10.20.1    110
1 ADo 10.10.11.0/30         10.10.30.1    110
2 ADo 192.168.10.0/24        10.10.30.1    110
3 ADo 192.168.20.0/24        10.10.30.1    110
4 ADo 192.168.30.0/24        10.10.30.1    110
5 ADo 192.168.40.0/24        10.10.30.1    110
6 ADo 192.168.50.0/24        10.10.20.1    110
[admin@MikroTik] > 
```

```
Switch x vIOS x VPC6 x VPC5 x VPC8 x VPC7 x NE40E x VPC9 x Mikrotik x

Error: Unrecognized command found at '^' position.
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
.....
public_Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.30.0/30     OSPF   10   110   D  10.10.11.1    Ethernet1/0/4
10.10.10.0/30     OSPF   10   110   D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24   OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24   OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24   OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24   OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.50.0/24   OSPF   10   11    D  10.10.11.1    Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.10.0/30     OSPF   10   10    10.10.10.2    Ethernet1/0/0
10.10.11.0/30     OSPF   10   10    10.10.11.2    Ethernet1/0/4
10.10.20.0/30     OSPF   10   100   10.10.20.1    Ethernet1/0/3
192.168.50.0/24   OSPF   10   1     192.168.50.1  Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
```

1.5 Praktikum 5: Pengujian Failover OSPF

1. Melakukan pengecekan routing pada kondisi normal ketika kedua link Cisco-Huawei aktif.
2. Melakukan pengujian jalur menggunakan perintah:

```
trace 192.168.50.10
```


3. Menonaktifkan salah satu link Cisco-Huawei menggunakan perintah shutdown.

```
* 10.10.10.2, from 2.2.2.2, 00:08:00, metric 11, traffic share count is 1
Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
 0 10.10.10.2 2 msec
 1 10.10.10.2 2 msec
 2 192.168.50.10 6 msec 3 msec 13 msec
```

4. Melakukan pengecekan ulang tabel routing OSPF.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O 10.10.10.0/30 [110/210] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
O 10.10.20.0/30 [110/200] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
C 10.10.30.0/30 is directly connected, GigabitEthernet0/1
L 10.10.30.1/32 is directly connected, GigabitEthernet0/1
C 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
L 192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
C 192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
L 192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
Router#
```

5. Melakukan trace kembali untuk memastikan lalu lintas tetap melewati jalur utama yang tersedia.

```
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
 0 10.10.10.2 2 msec
 1 10.10.10.2 2 msec
 2 192.168.50.10 6 msec 3 msec 13 msec
```

6. Menonaktifkan link Cisco-Huawei yang kedua sehingga kedua jalur utama terputus.

7. Melakukan pengecekan tabel routing untuk memastikan OSPF melakukan konvergensi.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O 10.10.10.0/30 [110/210] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
O 10.10.20.0/30 [110/200] via 10.10.30.2, 00:00:00, GigabitEthernet0/1
C 10.10.30.0/30 is directly connected, GigabitEthernet0/1
L 10.10.30.1/32 is directly connected, GigabitEthernet0/1
C 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
L 192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
C 192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
L 192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
Router#
```

8. Melakukan trace kembali dan mengamati perpindahan jalur melalui MikroTik sebagai backup path.

```

192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:17 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:17 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#trace 192.168.50.0
Type escape sequence to abort.
Tracing the route to 192.168.50.0
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.30.2 2 msec 1 msec 1 msec
  2 10.10.20.1 2 msec 2 msec 2 msec
  3 * * *
  4 * * *
  5 * * *
  6 * *

```

9. Mengaktifkan kembali kedua link Cisco-Huawei menggunakan perintah no shutdown.
10. Melakukan verifikasi bahwa jalur routing kembali menggunakan link utama Cisco-Huawei.

```

Switch * vIOS * VPC6 * VPC5 * VPC8 * VPC7 * NE40E * VPC9 * Mikrotik *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Router#show ip route 192.168.50.10
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:00:08 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:00:08 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:00:08 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip 192.168.50.0
^
% Invalid input detected at '^' marker.
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:00:19 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:00:19 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:00:19 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.10.2 2 msec
  10.10.11.2 3 msec
  10.10.10.2 2 msec
  2 192.168.50.10 8 msec 3 msec 3 msec
Router#

```

2 Analisis Hasil Percobaan

2.1 Praktikum 1: Konfigurasi Cisco Switch

Pada praktikum pertama dilakukan konfigurasi VLAN, access port, dan trunk port pada Cisco Switch. Berdasarkan hasil verifikasi menggunakan perintah `show vlan brief`, VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing) berhasil dibuat dan berada dalam status aktif. Selain itu, port-port yang terhubung ke masing-masing PC telah masuk ke VLAN yang sesuai sehingga segmentasi jaringan dapat diterapkan dengan baik.

Hasil verifikasi menggunakan perintah `show interfaces trunk` menunjukkan bahwa interface trunk berhasil membawa VLAN 10, 20, 30, dan 40 menuju Cisco Router. Hal ini sesuai dengan teori VLAN dan trunking IEEE 802.1Q, di mana satu link fisik dapat membawa beberapa VLAN sekaligus melalui proses tagging frame.

Keberhasilan konfigurasi dibuktikan dengan kemampuan perangkat dalam VLAN yang sama untuk saling berkomunikasi, sedangkan komunikasi antar VLAN belum dapat dilakukan karena belum terdapat perangkat Layer 3 yang melakukan routing. Hasil ini sesuai dengan teori bahwa VLAN secara default memisahkan broadcast domain sehingga perangkat pada VLAN yang berbeda tidak dapat saling berkomunikasi tanpa inter-VLAN routing.

Kendala yang mungkin terjadi selama praktikum adalah kesalahan penempatan access port pada VLAN yang tidak sesuai, kesalahan konfigurasi trunk, atau VLAN yang belum diizinkan melewati trunk.

Secara keseluruhan, tujuan praktikum berhasil dicapai karena segmentasi jaringan menggunakan VLAN dapat berfungsi sesuai rancangan.

2.2 Praktikum 2: Konfigurasi Cisco Router

Pada praktikum kedua dilakukan implementasi Router-on-a-Stick (ROAS) menggunakan Cisco Router sebagai gateway seluruh VLAN sekaligus DHCP Server untuk VLAN 10 dan VLAN 20. Hasil verifikasi menunjukkan bahwa seluruh subinterface berhasil dibuat dengan alamat IP yang sesuai sebagai gateway masing-masing VLAN.

PC pada VLAN 10 dan VLAN 20 berhasil memperoleh alamat IP secara otomatis melalui DHCP. Hal ini membuktikan bahwa layanan DHCP berjalan dengan baik dan router mampu mendistribusikan konfigurasi jaringan kepada client. Sementara itu, VLAN 30 dan VLAN 40 menggunakan alamat IP statis sesuai ketentuan praktikum.

Pengujian konektivitas menunjukkan bahwa seluruh client berhasil melakukan ping ke gateway masing-masing. Selain itu, pengujian antar VLAN juga berhasil dilakukan. Hasil ini sesuai dengan teori inter-VLAN routing, di mana router berfungsi sebagai perangkat Layer 3 yang meneruskan paket antar jaringan yang berbeda.

Keberhasilan praktikum menunjukkan bahwa konsep ROAS berhasil diterapkan. Faktor yang dapat menyebabkan kegagalan konfigurasi antara lain kesalahan VLAN ID pada subinterface, kesalahan alamat gateway, atau konfigurasi trunk yang tidak membawa VLAN tertentu. Secara keseluruhan, seluruh tujuan praktikum telah tercapai karena komunikasi antar VLAN berhasil diwujudkan melalui Cisco Router.

2.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antar Router

Pada praktikum ketiga dilakukan konfigurasi alamat IP pada Huawei Router, Cisco Router, dan Mikro-Tik Router untuk membentuk konektivitas antar perangkat jaringan. Setiap link point-to-point menggunakan subnet /30 sehingga penggunaan alamat IP menjadi lebih efisien sesuai teori pengalamatan jaringan.

Hasil pengujian menunjukkan bahwa seluruh interface berada pada kondisi *up* dan setiap router dapat melakukan ping ke router tetangga secara langsung. Selain itu, PC pada LAN Huawei berhasil melakukan komunikasi dengan gateway Huawei Router menggunakan jaringan 192.168.50.0/24.

Keberhasilan pengujian ping menunjukkan bahwa konfigurasi IP address dan aktivasi interface telah dilakukan dengan benar. Hal ini sesuai dengan teori bahwa komunikasi jaringan hanya dapat berlangsung apabila perangkat berada pada subnet yang sesuai dan memiliki konektivitas fisik yang aktif.

Permasalahan yang berpotensi muncul pada tahap ini adalah kesalahan subnet mask, penggunaan alamat IP yang tidak sesuai tabel addressing, atau interface yang belum diaktifkan menggunakan perintah `no shutdown` atau `undo shutdown`. Berdasarkan hasil yang diperoleh, seluruh koneksi antar router berhasil dibangun sehingga tahap ini dapat dinyatakan berhasil.

2.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

Pada praktikum keempat dilakukan implementasi protokol routing dinamis OSPF pada perangkat Cisco, Huawei, dan MikroTik. Tujuan utama praktikum ini adalah membangun pertukaran informasi

routing secara otomatis antar perangkat yang berasal dari vendor berbeda.

Hasil verifikasi menunjukkan bahwa seluruh router berhasil membentuk hubungan neighbour OSPF. Cisco Router berhasil mendeteksi Huawei dan MikroTik sebagai neighbour, Huawei Router berhasil membentuk peer dengan Cisco dan MikroTik, sedangkan MikroTik juga berhasil mengenali kedua router tersebut. Kondisi ini menunjukkan bahwa proses pertukaran Hello Packet dan pembentukan adjacency OSPF berjalan dengan baik.

Selain terbentuknya neighbour, tabel routing pada masing-masing perangkat berhasil memperoleh route dari jaringan lain secara dinamis. Cisco Router memperoleh route menuju LAN Huawei, Huawei memperoleh route menuju VLAN pada Cisco, dan MikroTik memperoleh seluruh informasi jaringan yang diiklankan oleh kedua router lainnya.

Pengujian end-to-end menunjukkan bahwa seluruh PC VLAN pada sisi Cisco dapat melakukan ping ke LAN Huawei, dan sebaliknya. Hasil tersebut sesuai dengan teori OSPF yang menggunakan algoritma Shortest Path First (SPF) untuk menentukan jalur terbaik berdasarkan nilai cost yang diberikan pada setiap interface.

Faktor yang dapat memengaruhi hasil praktikum antara lain ketidaksesuaian area OSPF, kesalahan network advertisement, router ID yang tidak unik, atau nilai cost yang tidak sesuai rancangan. Secara keseluruhan, implementasi OSPF multivendor berhasil dilakukan dan seluruh jaringan dapat saling berkomunikasi melalui routing dinamis.

2.5 Praktikum 5: Pengujian Failover OSPF

Pada praktikum kelima dilakukan pengujian mekanisme failover OSPF untuk mengetahui kemampuan jaringan dalam mempertahankan konektivitas ketika terjadi gangguan pada jalur utama. Pengujian dilakukan dalam tiga kondisi, yaitu saat seluruh link aktif, saat satu link utama mati, dan saat kedua link utama mati.

Pada kondisi normal, hasil *trace* menunjukkan bahwa trafik melewati jalur langsung Cisco-Huawei karena memiliki nilai cost paling rendah. Ketika salah satu link Cisco-Huawei dimatikan, trafik tetap dapat berjalan melalui link utama yang masih aktif sehingga konektivitas jaringan tidak terganggu. Hal ini menunjukkan adanya redundansi jalur pada topologi yang digunakan.

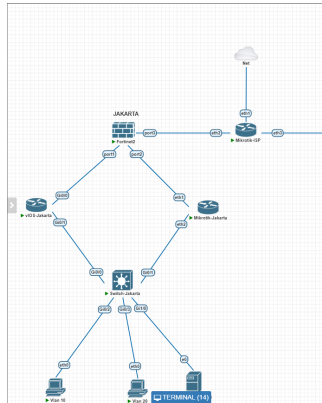
Saat kedua link utama Cisco-Huawei dimatikan, OSPF secara otomatis menghitung ulang jalur terbaik dan mengalihkan trafik melalui jalur cadangan yang melewati MikroTik. Proses ini membuktikan bahwa mekanisme konvergensi OSPF bekerja sesuai teori, yaitu mencari jalur alternatif yang masih tersedia ketika jalur utama mengalami kegagalan.

Setelah kedua link utama diaktifkan kembali, tabel routing kembali memilih jalur Cisco-Huawei karena memiliki cost yang lebih rendah dibandingkan jalur melalui MikroTik. Hasil ini menunjukkan bahwa pemilihan jalur OSPF sepenuhnya mengikuti perhitungan metric yang telah dikonfigurasi sebelumnya.

Beberapa faktor yang dapat memengaruhi hasil pengujian failover adalah waktu konvergensi OSPF, kestabilan koneksi antar perangkat, dan ketepatan konfigurasi cost pada setiap interface. Secara keseluruhan, praktikum berhasil membuktikan bahwa OSPF mampu menyediakan redundansi jaringan dan menjaga ketersediaan layanan ketika terjadi gangguan pada jalur utama.

3 Hasil Tugas Modul

1. Tugas modul 1



Gambar 1: 1.1

```
Switch>show vlan br

Press RETURN to get started.

*****
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* simulation. IDW is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IDW software or documentation to any third party for any *
* purpose is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Switch>show vlan br

VLAN Name                Status Ports
-----
1 default                active  Gi1/1, Gi1/2, Gi1/3
10 FINANCE               active  Gi0/2
20 IT                    active  Gi0/3
30 SERVER-HQ             active  Gi0/5
1002 FDDI-default         act/unsup
1003 token-ring-default   act/unsup
1004 Fddinet-default       act/unsup
1005 trnet-default         act/unsup

Switch>
```

Gambar 2: Enter Caption

```
Switch>show interfaces trunk

*****
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* purpose is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Switch>show interfaces trunk

VLAN Name                Status Ports
-----
1 default                active  Gi1/1, Gi1/2, Gi1/3
10 FINANCE               active  Gi0/2
20 IT                    active  Gi0/3
30 SERVER-HQ             active  Gi0/5
1002 FDDI-default         act/unsup
1003 token-ring-default   act/unsup
1004 Fddinet-default       act/unsup
1005 trnet-default         act/unsup

Switch>show interfaces trunk

Port      Mode      Encapsulation  Status  Native vlan
-----
Gi0/9     on        802.1Q         trunking  1
Gi0/1     on        802.1Q         trunking  1

Switch>show interfaces trunk

Port      Vlans allowed on trunk
-----
Gi0/9     10,20,60
Gi0/1     10,20,60

Switch>show interfaces trunk

Port      Vlans allowed and active in management domain
-----
Gi0/9     10,20,60
Gi0/1     10,20,60

Switch>show interfaces trunk

Port      Vlans in spanning tree forwarding state and not pruned
-----
Gi0/9     10,20,60
Gi0/1     10,20,60

Switch>
```

Gambar 3: 1.3

2. Tugas modul 2

```
Router>show ip interface brief

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* of the IDW software or documentation to any third party for any *
* purpose is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Router>show ip interface brief

Interface  IP-Address  OK? Method Status  P
-----
GigabitEthernet0/0  10.10.100.2  YES manual up      0
GigabitEthernet0/1  unassigned  YES unset  up      0
GigabitEthernet0/10 192.168.10.2 YES manual up      0
GigabitEthernet0/10 192.168.10.2 YES manual up      0
GigabitEthernet0/10 192.168.10.2 YES manual up      0
GigabitEthernet0/10 192.168.10.2 YES manual up      0
GigabitEthernet0/2  unassigned  YES unset  administratively down do
GigabitEthernet0/3  unassigned  YES unset  administratively down do
Router>
```

Gambar 4: 2.1

```

Router#show ip interface brief
Interface IP-Address OK? Method Status P
GigabitEthernet0/8 10.10.100.2 YES manual up
GigabitEthernet0/1 unassigned YES unset up
GigabitEthernet0/1.10 102.168.10.2 YES manual up
GigabitEthernet0/1.20 102.168.20.2 YES manual up
GigabitEthernet0/1.60 102.168.60.2 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down do
GigabitEthernet0/3 unassigned YES unset administratively down do
Router#show vrrp brief
Interface Grp Pri Tm State Master addr Group addr
Gig0/1.10 10 120 3531 Y Master 102.168.10.2 102.168.10.1
Gig0/1.20 20 90 3640 Y Backup 102.168.20.3 102.168.20.1
Gig0/1.60 60 120 3531 Y Master 102.168.60.2 102.168.60.1
Router#

```

Gambar 5: 2.2

```

Switch-Jakarta x Switch-Sura... x VIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vlan... x
Mikrotik-Sur... x Fortinet3 x Vlan 30 x Vlan 40 x Tinycore-Vla... x

!
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 10 ip 192.168.10.1
 vrrp 10 priority 120
end

Router#show running-config interface gi0/1.20
Building configuration...

Current configuration : 181 bytes
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 20 ip 192.168.20.1
 vrrp 20 priority 90
end

Router#show running-config interface gi0/1.60
Building configuration...

Current configuration : 149 bytes
!
interface GigabitEthernet0/1.60
 encapsulation dot1Q 60
 ip address 192.168.60.2 255.255.255.0
 vrrp 60 ip 192.168.60.1
 vrrp 60 priority 120
end

```

Gambar 6: 2.3

```

Router#ping 10.10.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router#

```

Gambar 7: 2.4

3. Tugas modul 3

```

Switch-Jakarta x Switch-Sura... x VIOS-Jakarta x Ubuntu-Vlan... x Mikrotik-Jak... x Fortinet3 x
Mikrotik-Sur... x Fortinet3 x Vlan 30 x Vlan 40 x Tinycore-Vla... x

addip v10/11/13 > /ip address print
# ADDRESS NETWORK INTERFACE
0 192.168.10.2/24 102.168.10.0 vlan10-eth0
1 192.168.20.2/24 102.168.20.0 vlan20-eth0
2 192.168.60.2/24 102.168.60.0 vlan60-eth0
3 10.10.101.2/24 10.10.101.0 eth0/1
4 192.168.10.1/24 102.168.10.1 vrrp10
5 192.168.20.1/24 102.168.20.1 vrrp20
6 192.168.60.1/24 102.168.60.1 vrrp60

```

Gambar 8: 3.1

```
admin@mikrotik:~$ /interface vrrp print
Flags: K - disabled, I - invalid, R - running, M - master, B - backup
#  NAME      STATE  VRRP-ADDRESS  VRRP-PRI  INTERFACE  V V3...
0  vrrp10    vlan...  192.168.10.10  10  vrrp10     2  ip4
1  vrrp20    vlan...  192.168.20.10  10  vrrp20     2  ip4
2  vrrp60    vlan...  192.168.60.10  10  vrrp60     2  ip4
```

Gambar 9: 3.2

```
admin@mikrotik:~$ /ip dhcp-relay print
Flags: K - disabled, I - invalid
#  NAME      INTERFACE  DHCP-SERVER  LOCAL-ADDRESS
0  relay1    vrrp10-finance  192.168.60.10  192.168.10.3
1  relay2    vrrp20-finance  192.168.60.10  192.168.20.3
```

Gambar 10: 3.3

```
admin@mikrotik:~$ /ip route print
Flags: K - disabled, A - active, D - dynamic,
#  ADDRESS  PREFIX  LENGTH  OUT-INTERFACE  DISTANCE
0  A 5 8.8.8.8/8  8  10.10.10.1  1
1  A 0 0.0.0.0/0  0  10.10.10.1  1
2  A 0 0.0.0.0/0  0  10.10.10.1  1
3  A 0 0.0.0.0/0  0  10.10.10.1  1
4  A 0 0.0.0.0/0  0  10.10.10.1  1
5  A 0 0.0.0.0/0  0  10.10.10.1  1
```

Gambar 11: 3.4

```
[admin@mikrotik] > ping 10.10.101.1
SEQ HOST      SIZE TTL TIME STATUS
0 10.10.101.1  56 255 3ms
1 10.10.101.1  56 255 0ms
2 10.10.101.1  56 255 0ms
3 10.10.101.1  56 255 0ms
4 10.10.101.1  56 255 0ms
5 10.10.101.1  56 255 0ms
sent=6 received=6 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=3ms
```

Gambar 12: 3.5

4. Tugas modul 4

```
admin@mikrotik:~$ /ip dhcp-relay print
Flags: K - disabled, I - invalid
#  NAME      INTERFACE  DHCP-SERVER  LOCAL-ADDRESS
0  relay1    vrrp10-finance  192.168.60.10  192.168.10.3
1  relay2    vrrp20-finance  192.168.60.10  192.168.20.3
```

Gambar 13: 4.1

```
admin@mikrotik:~$ /ip route print
Flags: K - disabled, A - active, D - dynamic,
#  ADDRESS  PREFIX  LENGTH  OUT-INTERFACE  DISTANCE
0  A 5 8.8.8.8/8  8  10.10.10.1  1
1  A 0 0.0.0.0/0  0  10.10.10.1  1
2  A 0 0.0.0.0/0  0  10.10.10.1  1
3  A 0 0.0.0.0/0  0  10.10.10.1  1
4  A 0 0.0.0.0/0  0  10.10.10.1  1
5  A 0 0.0.0.0/0  0  10.10.10.1  1
```

Gambar 14: 4.2

```
root@kvm:~# cat /etc/dhcp/dhcpd.conf
authoritative;

default-lease-time 600;
max-lease-time 7200;

option domain-name-servers 8.8.8.8, 1.1.1.1;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.10.255;
}

subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.200;
    option routers 192.168.20.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.20.255;
}

subnet 192.168.60.0 netmask 255.255.255.0 {
    option routers 192.168.60.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.60.255;
}

root@kvm:~#
```

Gambar 15: 4.3

```
root@kvm:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 64 bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=25.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=25.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=110 time=25.0 ms
^C
- 8.8.8.8 ping statistics -
3 packets transmitted, 3 received, 0% packet loss, time 200ms
rtt min/avg/max/mdev = 25.579/25.812/26.047/0.191 ms
root@kvm:~#
```

Gambar 16: 4.4

```
edit 1
set name "LAN-TO-INTERNET"
set uuid 796b6012-0504-51f1-87bf-5abed7acce
set strictf "port1" "port2"
set strictf "port3"
set action accept
set reader "all"
set disabler "all"
set schedule "always"
set service "ALL"
set not enable
next
edit 2
set name "LAN-to-GRE"
set uuid 48ba521c-0509-51f1-84ce-04c190359057
set strictf "port1" "port2"
set strictf "GRE-JKT-SBY"
set action accept
set reader "all"
set disabler "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "GRE-to-LAN"
set uuid 1892c858-0509-51f1-f5a6-51dd8fa54d8a
set strictf "GRE-JKT-SBY"
set strictf "port1" "port2"
set action accept
set reader "all"
set disabler "all"
set schedule "always"
set service "ALL"
next
```

Gambar 19: 5.3

5. Tugas modul 5

```
portinfo-VN04-KVM # show
portinfo-VN04-KVM # show
portinfo-VN04-KVM # get system interface physical
[show]
==[port1]==
mode: static
ip: 10.10.100.1 255.255.255.252
link: 1/0
status: up
speed: 100Mbps (Duplex: full)
PFC: none
PFC cap: none
==[port2]==
mode: static
ip: 10.10.101.1 255.255.255.252
link: 1/0
status: up
speed: 100Mbps (Duplex: full)
PFC: none
PFC cap: none
==[port3]==
mode: static
ip: 10.0.0.2 255.255.255.252
link: 1/0
status: up
speed: 100Mbps (Duplex: full)
PFC: none
```

Gambar 17: 5.1

```
portinfo-VN04-KVM # get router info routing-table all
router: 0 - Vlan0/0, connected 1 - static, R - RDP, W - OSPF
0 - OSPF, IA - OSPF inter area
R2 - OSPF NSSA external type 2, R3 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS IS, L1 - IS IS level-1, L2 - IS IS level-2, ia - IS IS inter are
V - BGP VRFid
* - candidate default
Routing table for VRF=0
V
0.0.0.0/0 (0/0) via 10.0.0.2, port3 (1/0)
10.0.0.0/8 is directly connected, port3
10.10.100.0/24 is directly connected, port1
10.10.101.0/24 is directly connected, port2
172.16.0.0/24 is directly connected, GRE-JKT-SBY
172.16.0.2/32 is directly connected, GRE-JKT-SBY
192.168.16.0/24 (1/0) via 10.10.100.2, port1 (1/0)
192.168.16.0/24 (1/0) via 10.10.101.2, port2 (1/0)
192.168.20.0/24 (1/0) via 172.16.0.2, GRE-JKT-SBY, 01:31:30, [1/0]
E2 192.168.40.0/24 (110/10) via 172.16.0.2, GRE-JKT-SBY, 01:31:30, [1/0]
E2 192.168.40.0/24 (110/10) via 172.16.0.2, GRE-JKT-SBY, 01:31:30, [1/0]
192.168.40.0/24 (110/10) via 10.10.100.2, port1 (1/0)
192.168.40.0/24 (110/10) via 10.10.101.2, port2 (1/0)
```

Gambar 18: 5.2

```
portinfo-VN04-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=22.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=23.0 ms
^C
-- 8.8.8.8 ping statistics --
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 22.3/22.6/23.0 ms
```

Gambar 20: 5.4

```
portinfo-VN04-KVM # execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=2.9 ms
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.6 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
-- 172.16.0.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.9/2.9 ms
```

Gambar 21: 5.5

```
portinfo-VN04-KVM # get router info ospf neighbor
ospf process 0, VRF 0
Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 Full/- 00:00:37 172.16.0.2 GRE-JKT-SBY
```

Gambar 22: 5.6

```
portinfo-VN04-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2 192.168.40.0/24 (110/10) via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
0 E2 192.168.40.0/24 (110/10) via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
```

Gambar 23: 5.7

Gambar 31: 7.2

Gambar 32: 7.3

Gambar 33: 7.4

Gambar 34: 7.5

Gambar 35: 7.6

Gambar 36: 7.7

Gambar 37: 7.8

8. Tugas modul 8

Gambar 39: 8.2

```
FortiGate-VN64-KVM # Timeout

FortiGate-VN64-KVM login: admin
Password:
Welcome!

FortiGate-VN64-KVM # get system interface physical
== [onboard]
==[port1]
mode: static
ip: 10.0.13.2 255.255.255.252
ipv6: ::/0
status: up
speed: 1000Mbps (Duplex: full)
FEC: none
FEC_cap: none
==[port2]
mode: static
ip: 10.10.200.1 255.255.255.252
ipv6: ::/0
status: up
speed: 1000Mbps (Duplex: full)
FEC: none
FEC_cap: none
==[port3]
mode: static
ip: 0.0.0.0 0.0.0.0
ipv6: ::/0
status: down
speed: n/a
FEC: none
--More--
```

Gambar 38: 8.1

```
config firewall policy
edit 1
set name "LAN-TO-INTERNET"
set uid 1c3c4b6-6503-51f1-2b6b-03775fd08f42
set srcintf "port2"
set dstintf "port1"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN-to-GRE"
set uid 447f06eb-6509-51f1-25b6-83c7a93f26af
set srcintf "port2"
set dstintf "GRE-SBY-JKT"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "GRE-to-LAN"
set uid 448c77d2-6509-51f1-a24f-2ca1b8f7fb82
set srcintf "GRE-SBY-JKT"
set dstintf "port2"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
```

Gambar 40: 8.3

```
FortiGate-VN64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=23.3 ms
^C
-- 8.8.8.8 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 22.2/22.8/23.3 ms
```

Gambar 41: 8.4

```
FortiGate-VN64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=1.2 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.3 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.2 ms
^C
-- 172.16.0.1 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/1.1/1.2 ms
```

Gambar 42: 8.5

```
FortiGate-VN64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID Pri State Dead Time Address Interface
1.1.1.1 1 Full/- 00:00:31 172.16.0.1 GRE-SBY-JKT
```

Gambar 43: 8.6

```
FortiGate-VN64-KVM # get router info routing-table ospf
Routing table for VN64:
E2 192.168.16.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
E2 192.168.28.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
E2 192.168.68.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
```

Gambar 44: 8.7

9. Tugas Modul 9

```
FortiGate-VN64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.3 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.6 ms
^C
-- 10.0.12.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms
```

Gambar 45: 9.1.1

```

FortiGate-VN04-KVM # execute ping 10.0.13.2
PING 10.0.13.2 (10.0.13.2): 56 data bytes
64 bytes from 10.0.13.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.13.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.13.2: icmp_seq=2 ttl=254 time=1.1 ms
^C
--- 10.0.13.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms

```

Gambar 46: 9.1.2

```

FortiGate-VN04-KVM # execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
64 bytes from 172.16.0.2: icmp_seq=0 ttl=255 time=3.9 ms
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/2.2/3.9 ms

```

Gambar 47: 9.2.1

```

FortiGate-VN04-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=4.4 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.4 ms
^C
--- 172.16.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.4/2.4/4.4 ms

```

Gambar 48: 9.2.2

```

FortiGate-VN04-KVM # get router info ospf neighbor
OSPF process 0, VRF all
Neighbor ID      Pri  State          Dead Time Address          Interface
2.2.2.2          1  Full/ -        00:00:37    172.16.0.2        GRE-KT-SRV

```

Gambar 49: 9.3.1

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VRF all
O E2  192.168.16.0/24 (110/10) via 172.16.0.2, GRE-KT-SRV, 01:34:16, [1/0]
O E2  192.168.40.0/24 (110/10) via 172.16.0.2, GRE-KT-SRV, 01:34:16, [1/0]

```

Gambar 50: 9.4.1

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VRF all
O E2  192.168.16.0/24 (110/10) via 172.16.0.1, GRE-SRV-KT, 01:47:32, [1/0]
O E2  192.168.20.0/24 (110/10) via 172.16.0.1, GRE-SRV-KT, 01:47:32, [1/0]
O E2  192.168.40.0/24 (110/10) via 172.16.0.1, GRE-SRV-KT, 01:47:32, [1/0]

```

Gambar 51: 9.4.2

```

RPS> ping 192.168.40.10
64 bytes from 192.168.40.10: icmp_seq=1 ttl=60 time=13.004 ms
64 bytes from 192.168.40.10: icmp_seq=2 ttl=60 time=11.706 ms
64 bytes from 192.168.40.10: icmp_seq=3 ttl=60 time=10.312 ms
^C
RPS>

```

Gambar 52: 9.5

```

RPS> ping 192.168.10.100
64 bytes from 192.168.10.100: icmp_seq=1 ttl=60 time=12.802 ms
64 bytes from 192.168.10.100: icmp_seq=2 ttl=60 time=7.930 ms
64 bytes from 192.168.10.100: icmp_seq=3 ttl=60 time=7.014 ms
^C

```

Gambar 53: 9.6

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VRF all
O E2  192.168.16.0/24 (110/10) via 172.16.0.2, GRE-KT-SRV, 01:34:16, [1/0]
O E2  192.168.40.0/24 (110/10) via 172.16.0.2, GRE-KT-SRV, 01:34:16, [1/0]

```

Gambar 54: 9.7.1

```

FortiGate-VN04-KVM # get router info routing-table ospf
Routing table for VRF all
O E2  192.168.16.0/24 (110/10) via 172.16.0.1, GRE-SRV-KT, 01:47:32, [1/0]
O E2  192.168.20.0/24 (110/10) via 172.16.0.1, GRE-SRV-KT, 01:47:32, [1/0]
O E2  192.168.40.0/24 (110/10) via 172.16.0.1, GRE-SRV-KT, 01:47:32, [1/0]

```

Gambar 55: 9.7.2

```
VPCS> ip dhcp
DDORA IP 192.168.10.102/24 GW 192.168.10.1
```

Gambar 56: 10.1

```
VPCS> ip dhcp
DORA IP 192.168.30.200/24 GW 192.168.30.1
```

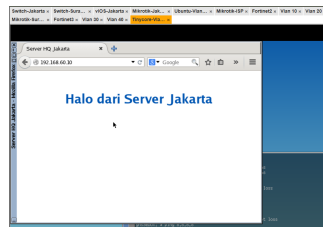
Gambar 57: 10.2

```
VPCS> ping 8.8.8.8
64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=20.083 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=25.015 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=20.961 ms
^C
VPCS>
```

Gambar 58: 10.4

```
VPCS> ping 192.168.40.10
64 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=8.299 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=4.425 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.837 ms
^C
VPCS>
```

Gambar 59: 10.5



Gambar 60: 10.6

```
writeable vty0-157 no & get router info ospf neighbor
OSPF process 0, Ver 81
Neighbor ID Pri State Dead Time Address Interface
1.1.1.1 1 Full/ 00:00:31 172.16.0.1 G0E/0/0/0/0
```

Gambar 61: 10.7

```
VPCS> trace 192.168.40.10
Tracer to 192.168.40.10, 3 hops max, press Ctrl+C to stop
 1 192.168.10.2 11.530 ms 2.103 ms 4.478 ms
 2 19.10.100.1 0.112 ms 0.102 ms 0.364 ms
 3 172.16.0.2 5.450 ms 5.500 ms 11.021 ms
 4 10.0.0.200.2 5.802 ms 5.912 ms 4.761 ms
 5 *192.168.40.10 11.520 ms (ICMP type3, code3, Destination port unreachable)
^C
```

Gambar 62: 10.8

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, seluruh tujuan praktikum pada Modul 5 berhasil dicapai. Pada Praktikum 1, konfigurasi VLAN, access port, dan trunk port pada Cisco Switch berhasil diterapkan sehingga segmentasi jaringan dapat berjalan sesuai rancangan. Pada Praktikum 2, implementasi Router-on-a-Stick (ROAS) berhasil memungkinkan komunikasi antar VLAN serta pemberian alamat IP secara otomatis melalui layanan DHCP pada Cisco Router.

Pada Praktikum 3, konfigurasi alamat IP antar perangkat Cisco, Huawei, dan MikroTik berhasil dilakukan sehingga seluruh router dapat saling terhubung dan berkomunikasi melalui jaringan point-to-point. Selanjutnya, pada Praktikum 4, protokol routing dinamis OSPF berhasil dikonfigurasi pada

perangkat multivendor sehingga seluruh jaringan dapat saling bertukar informasi routing dan melakukan komunikasi end-to-end. Pada Praktikum 5, mekanisme failover OSPF berhasil diuji, di mana jalur cadangan secara otomatis digunakan ketika jalur utama mengalami gangguan dan kembali menggunakan jalur utama setelah koneksi dipulihkan.

Hasil yang diperoleh selama praktikum sesuai dengan teori yang telah dipelajari mengenai VLAN, inter-VLAN routing, pengalamatan IP, routing dinamis OSPF, serta konsep redundansi jaringan menggunakan failover. OSPF terbukti mampu memilih jalur terbaik berdasarkan nilai cost dan melakukan konvergensi secara otomatis saat terjadi perubahan topologi jaringan.

Melalui praktikum ini, praktikan memperoleh pemahaman yang lebih mendalam mengenai proses perancangan, konfigurasi, verifikasi, dan troubleshooting jaringan komputer pada lingkungan multivendor. Selain itu, praktikan juga memahami pentingnya perencanaan addressing, pengelolaan VLAN, penggunaan routing dinamis, serta penerapan redundansi jaringan untuk meningkatkan keandalan dan ketersediaan layanan jaringan.

5 Lampiran



Gambar 63: foto